

ORGANIC CHEMISTRY: MASTER TEXTBOOK

1. Introduction to Organic Chemistry

Carbon & Bonding

Organic chemistry is the study of carbon-containing compounds. Carbon's ability to form stable covalent bonds with itself (catenation) and other elements is the basis of life.

Key concepts: Hybridization (sp , sp^2 , sp^3), Electronegativity, and Bond Polarity.

2. Structural Representation & Isomerism

Isomerism

Isomers are compounds with the same molecular formula but different arrangements of atoms.

- **Structural Isomerism**: Chain, Position, Functional group.
- **Stereoisomerism**: Geometric (cis-trans) and Optical (enantiomers).

3. Reaction Mechanisms

Fundamental Processes

Organic reactions follow specific pathways:

- **Nucleophilic Substitution (SN1/SN2)** Nucleophile replaces a leaving group.
- **Electrophilic Addition** Occurs across double/triple bonds.
- **Elimination (E1/E2)** Formation of unsaturated compounds.

4. Functional Groups

Classification

Reactivity is governed by functional groups such as Alcohols (-OH), Carboxylic Acids (-COOH), Aldehydes (-CHO), and Amines (-NH²).